

Phase A Methodology: A Causal Audit of Semantic Induction Heads on Single-hop In-context Relations

Daniella Simonovsky and Max Golubkov

Methodology specification — September 2026

1 Goal and Guiding Principles

Ren et al. [1] define *semantic induction heads* (SIH) observationally, via the Relation Index (RI), and report correlations with ICL emergence — with no interventions. Phase A asks: **do heads passing the SIH definition causally carry the model’s single-hop in-context relation ability, and is the definition complete — or are these heads the visible tip of a larger circuit?** (Phase B extends to two-hop composition.)

Four principles constrain everything below:

1. **Audit, not verification.** We measure the observational score *and* the causal effect on *all* $32 \times 32 = 1024$ heads, then compare the two rankings. Selecting heads by RI and testing only those would make the definition’s misses invisible by construction.
2. **Definitional fidelity.** Parameters inside the SIH definition (notably the dominance threshold $\tau = 2.2$) are kept exactly as published; we may not redesign the thing we are auditing. Deviations appear only as sensitivity analyses.
3. **Null-calibrated, pre-registered thresholds.** Every selection threshold is a *rule* fixed before seeing results, calibrated against an explicit null distribution — never a percentile of the empirical score distribution (which would “select” a fixed fraction of heads even under pure noise).
4. **Discovery / validation split.** All screening and selection uses the 89 discovery families (even family ids); the 87 validation families are touched only by the final faithfulness and generalization tests.

2 Experimental Substrate

Model. OLMo-2-1124-7B (base), revision pinned in `model_lock_olmo2.json`; fp16 across two 11–12 GB GPUs. Pythia-6.9B retained as a contrast model for later comparisons.

Data. `pilot_v3/data_singlehop/singlehop_v1_*.jsonl`: 200 kinship families (relation *mother-of* over fictitious names), each with three variants $\times$ two fact orders: *base*; *corrupted* — the two answer-side names swapped inside the fact lines, so the former distractor becomes the correct answer while every mention count is unchanged and the two prompts are **token-aligned position-by-position** (≤ 6 differing tokens); *reorder* — fact lines shuffled. Every fact line (demonstrations included) carries construction-time annotations: relation, head/tail entities, and their character spans — the anchors RI needs, with no parser noise (an improvement over [1]’s spaCy/AGENDA

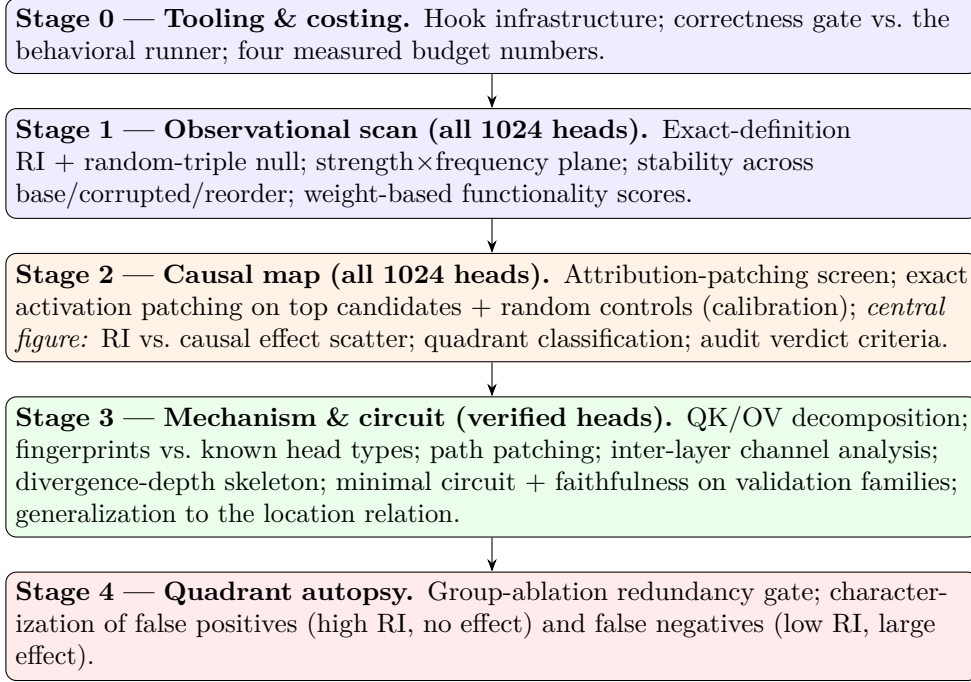


Figure 1: Phase-A pipeline. Stages 1–2 are computed on *every* head; stage 3 only on causally verified heads.

annotation). A separate 50-family *location* world (relation *inside*) is reserved for the stage-3 generalization test.

Working condition. 4-shot with single-hop-only demonstrations (behavioral baseline: base 0.93, corrupted 0.94, reorder 0.89 both-orders accuracy; working set of 176/200 families correct on base+corrupted in both orders). 0-shot is the secondary contrast. Demonstration sets are identical across all variants and orders of a family, so twins differ only in the test block.

Primary metric for all interventions. The candidate logit difference at the answer position,

$$M(x) = \text{logit}(y_{\text{gold}} | x) - \text{logit}(y_{\text{distractor}} | x), \quad (1)$$

following the recommendations of [6] (logit difference rather than probability or accuracy: linear in the final residual stream, sensitive, sign-interpretable). On a clean/corrupted twin the gold and distractor swap roles, so patching effects have a *signed* prediction (the difference should flip toward the corrupted run’s answer), not merely “performance drops”.

3 Stage 0 — Tooling and Costing

What. Stand up hook-based access to internal activations (read attention patterns and per-head outputs; write/replace head outputs mid-run) via TransformerLens [14], with raw PyTorch forward hooks on the HF model as fallback if the wrapper misbehaves on our sharded 2-GPU setup.

Correctness gate. Before any measurement: the hooked model must reproduce the behavioral runner’s candidate scores on 20 prompts within numerical tolerance. No gate, no science.

Why. All later budgets depend on four numbers that cannot be reliably guessed for this hardware.

#	Measurement	What it budgets
1	sec + GiB of a forward pass with attention caching	Stage-1 scan time ($\times$ #prompts); cache is $L H n^2 \approx 32 \cdot 32 \cdot 350^2 \approx 1.3 \times 10^8$ values/prompt $\Rightarrow$ streaming policy
2	sec of a full RI computation per prompt	Verifies the cheap order of operations (QK filter from cached attention first; OV vocabulary projection only for surviving (head, position) pairs at entity positions — naively this stage costs ~ 30 forwards/prompt)
3	sec of one forward with a hook intervention	Unit price of exact patching (stage-2 verification budget)
4	does a backward pass fit in GPU memory?	Gates attribution patching (needs $\partial M/\partial a$ for all sites); if not: 4-GPU jobs or memory-saving fallbacks

4 Stage 1 — Observational Scan (all heads)

Aim and scope. Identify candidates using RI and characterize what their scores measure, before assessing their causal role with other methods. The completed scan, `stage1_v4_calibrated`, covered all 1,024 heads of the pinned OLMo-2-1124-7B model on 534 prompts: 89 discovery families, three variants, and two fact orders. The extension below reuses this population; validation families and Stage-2 effects are not used for selection. It is implemented but awaits GPU completion and verification. This specification supersedes earlier Stage-1 planning summaries elsewhere in this document.

4.1 RI definition and original checks

For a fact $T = (t_s, r, t_o)$, the source is the child and the target is its mother. Anchor the source at its last token s ; score the target’s first and last tokens separately. For each position $j \geq \max(s, o_{\text{last}})$, require

$$s = \arg \max_{k \leq j} A_{j,k}^h, \quad \frac{A_{j,s}^h}{\max_{k \leq j, k \neq s} A_{j,k}^h + 10^{-9}} > 2.2. \quad (2)$$

This QK condition uses captured model attention and permits self-attention. For a passing event, project the raw current-token embedding:

$$p^{h,j} = \text{softmax}(e_j W_V^h W_O^h W_U), \quad q_t^{h,j} = \max \left(0, p_t^{h,j} - \frac{1}{|C_j|} \sum_{u \in C_j} p_u^{h,j} \right), \quad (3)$$

$$\text{RI}_{h,j}(t) = \frac{q_t^{h,j}}{\sum_{u \in C_j} q_u^{h,j}}, \quad (4)$$

where C_j contains unique token IDs visible through j . A zero denominator gives an undefined score. This follows the RI projection in [1], with explicit adaptations: names remain multi-token, function words remain, and QK uses the model’s contextual attention. It is not an exact replication of the paper’s semantic-relation preprocessing. The OV score is neither the actual contextual head output nor a causal logit change.

Before the original scan, mandatory checks covered behavioral-score replication on six rows from one family, exact inert-hook agreement, attention availability, and a one-prompt self-patch. Behavioral drift had to be below 0.05 and self-patch metric drift below 10^{-3} . Attention-capture logit drift was recorded without a numerical acceptance threshold. A small eight-head patching preview was

descriptive. Intervention checks used the first differing answer token after any shared answer prefix, with patches confined to original prompt positions. These limited checks were not full-dataset equivalence tests.

4.2 Test-only scope and comparison groups

The extension includes events only when *both the annotated fact and current position are in the test block*. Four demonstrations precede each test. Their sequence repeats across the six variants/orders of a family, and demonstrations are drawn from a shared bank of 16. Excluding demo events avoids directly pooling their repeated scores; it does not remove ICL conditioning. RI normalization still uses the full visible context, including demonstrations.

Two scopes are reported separately:

1. **Relation in a fact:** all four test facts, including distractors.
2. **Query-relevant fact:** only the fact whose source is named in the question and whose target is the correct answer for that variant.

Separate all-test positions, earlier test positions, and the final bare-prompt token, which predicts the first answer token. Within each group, distinguish self, previous-token, and longer-distance attention. These overlapping views are not additive.

For each scored event, compare the target with two control sets:

- **Alternative names:** all other fully visible fact-target names in the same test, without an equal-token-length restriction.
- **General words:** up to three unique alphabetic non-name word types from the visible test. Sample deterministically using seed 20260916 and the prompt ID/position; share the sample across heads and facts at that position. Use each word’s first fully visible occurrence. Exclude entity names and punctuation.

Compute separate first-token and last-token scores for targets and controls. Equal anchor-token IDs remain ties, not evidence of successful or failed entity discrimination. General words test preference above linguistic background; alternative names test specificity among entities. Neither comparison alone establishes semantic ability.

4.3 Event statistics, aggregation and variant consistency

For anchor a and available control set B_e , report the target score and

$$\Delta_{h,e}^{(a)} = \text{RI}_{h,e}(t_o^{(a)}) - \frac{1}{|B_e|} \sum_{b \in B_e} \text{RI}_{h,e}(b^{(a)}). \quad (5)$$

Also record the target’s rank interval, fractions of controls beaten or tied, and whether it ranks first with or without ties. An empty control set yields an undefined comparison. Preserve control identities, positions and scores.

Average each statistic within a family, then equally across families with defined observations for that statistic. Report its event count and active-family count. Target/control means used in a comparison share the same event population. Families without observations have undefined conditional strength, not RI zero. QK frequency is passes divided by eligible fact–position opportunities: report both the equal-family mean, including zero-pass families, and the pooled event-weighted ratio. All 1,024 heads remain in the tables.

Compare base with corrupted and reordered prompts by matching family, order and source-name identity, not fact index. Report changes in QK counts/frequencies and RI statistics, with missing passes explicit. Final-position comparisons align the prediction position; all-test means may include different passing positions. Variants are not independent samples. For fixed head, current token and visible token set, the raw-embedding OV vector is fixed: a changed target assignment can change the scored coordinate without changing that vector. Thus paired RI differences alone do not demonstrate contextual tracking.

4.4 Candidate selection and follow-up audit

Before computing the additional OV scores, fix a descriptive discovery rule: require at least 20 scored events across 10 active families for the statistic, then take the top 10 positive family means for each combination of

- target RI, alternative-name gap, or general-word gap;
- first or last target token;
- all facts or the query-relevant fact;
- all-test or final positions.

The shortlist is their union, with layer/head order breaking ranking ties. These are exploratory selection rules, informed by the earlier audit, not calibrated significance thresholds. Record every selection reason and retain unsupported heads in the full results. Historical candidates L3H11/L9H22 remain references, not forced selections.

After selection, inspect family and token concentration, removal of the most influential family, repeated current/target tokens, concrete examples, and first/last-token sensitivity. These planned diagnostics investigate why a head was selected; they do not independently validate semantics. Validation families remain reserved for later tests.

4.5 Historical calibration and sensitivity

Retain the original pooled RI as a reference. Its historical rule used the 99.9th percentile of random-token control means among heads with at least 50 scored events; it is descriptive, not a calibrated per-head test.

The completed matched-target analysis is a separate statistic: test–test events with a candidate fact target, both candidates fully visible, equal token lengths and distinct first tokens. It compared target/control family means using 100,000 family-consistent swaps, required 50 events across 10 families, and applied Holm correction across 1,024 heads. Its interpretation assumes conditional target/control exchangeability. The new, broader name/word comparisons do not inherit its calibration or p -values.

The original scan also retained all dominance ratios for which the annotated source was the attention argmax. Filtering saved events at their global 95th percentile, approximately 4.482, provided a completed sensitivity analysis without recalibrating the matched null. That population includes repeated demonstrations and is event-weighted; the percentile is not an independently justified optimal threshold. The extension retains $\tau = 2.2$.

4.6 Execution, verification and outputs

CPU preparation validates the original event counts, discovery population, annotations and token positions, and freezes code/input hashes and selection rules. GPU completion reuses saved QK

events and computes missing OV scores, including last-token alternative names and general-word controls. No new model forward pass is required. Token IDs/offsets must reproduce the saved prompts; recomputed historical scores must agree within absolute tolerance 2×10^{-4} and relative tolerance 2×10^{-3} . Failure stops the run. Verified per-head files support resumption.

CPU analysis produces event scores, all-head summaries, family statistics, paired-variant comparisons and the candidate list. Completion is recorded only after all expected events and output files are verified. The extension provides RI-based candidates and diagnostics; contextual-output measurements, causal effects and circuit tests belong to later stages.

5 Stage 2 — Full Causal Map (all heads)

5.1 Attribution-patching screen

Exact activation patching for all heads is infeasible (1024×178 twin pairs $\approx 1.8 \times 10^5$ forwards). Instead, a first-order estimate of every head’s patching effect is obtained from three passes per twin pair [7]:

$$\widehat{\Delta M}_h = (a_h^{\text{corr}} - a_h^{\text{clean}})^\top \frac{\partial M}{\partial a_h} \Big|_{\text{clean}}, \quad (6)$$

where $a_h \in \mathbb{R}^{d_{\text{model}}}$ is head h ’s output at the answer position. One clean forward, one corrupted forward, and one backward pass of M yield $\widehat{\Delta M}_h$ for *all* heads (and MLP layers, and all positions) simultaneously: ≈ 700 forward-equivalents for the full discovery set.

Known failure mode and mitigation. The estimate is a first-order Taylor approximation; it under-estimates precisely where the path from the activation to the metric crosses saturated softmax plateaus — i.e., it is biased *against* sharply-attending heads, our most interesting candidates. Therefore:

1. **Calibration (pre-registered):** exact activation patching on the union of the top-25 heads by each ranking (RI, attribution, weight scores) plus 25 random heads, on 40 twin pairs. The exact-vs-estimate agreement (Spearman correlation) on this sample is itself a reported result; if $\rho < 0.7$, the screen is upgraded to integrated-gradients attribution (EAP-IG, $m=5$ interpolation steps [8]) and re-calibrated.
2. Exact patching is always the measurement of record for any head individually discussed.

5.2 The central figure and the audit verdict

The main deliverable: a scatter of all 1024 heads, x = RI strength, y = causal effect (calibrated estimate; exact where available), color = weight-based copy score. Threshold-free statistics reported: Spearman rank correlation between RI and causal effect over all heads; overlap curves (top- k Jaccard).

Pre-registered verdict criteria (to be confirmed with the team before stage 2 runs): the definition is judged *sound* if RI-passing heads have systematically larger causal effects than layer-matched non-passing heads; *incomplete* if at least one head in the top decile of exact-patching effect lies below even the *lenient* (99th-percentile) RI null threshold — so that the incompleteness claim cannot be manufactured by our own strictness; *misleading* if the median causal effect of RI-passing heads does not exceed that of random heads. These are not mutually exclusive (e.g., sound and incomplete). Note that the selection threshold never gates *which heads are studied* — the stage-2 causal map covers all 1024 heads unconditionally; the threshold only names the set attributed to the definition, and the primary quantitative claims are threshold-free (rank correlation, overlap curves).

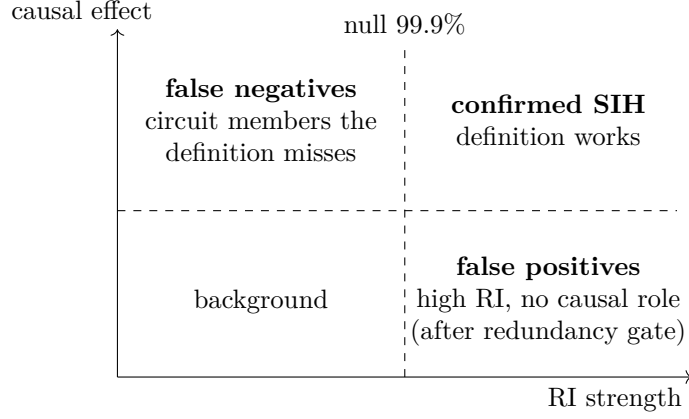


Figure 2: Quadrant reading of the central scatter. Stage 4 characterizes the two off-diagonal quadrants.

6 Stage 3 — Mechanism and Circuit (verified heads only)

Tools, in order of application:

1. **QK side — edge-following score.** Using the fact anchors: attention mass from the answer position (and from the question-entity mention) to the correct fact’s tail token, vs. positional controls; *signed* version on corrupted twins (does attention move to the swapped source?).
2. **OV side — what is written.** Project the head’s actual output at the answer position to the vocabulary; verify it promotes the specific answer name; spectral copying check on the name subspace [3].
3. **Fingerprints vs. known head types.** Standardized scores locate our heads in the existing taxonomy, answering “new kind, or re-labeled known kind?”: prefix-matching and copying scores on random repeated sequences [2]; retrieval-head score (fraction of copy events where the head’s top attention sits on the copied token) [10]; function-vector mediation (does the head’s mean output over relation prompts causally induce the relation zero-shot?) [11]. Deliverable: a head $\times$ fingerprint table.
4. **Path patching** [4, 5]: which upstream components feed the verified heads’ queries/keys/values, and which downstream components consume their output — the head-vs-circuit question at edge resolution. Position-aware edges follow [9] given our templated prompts.
5. **Inter-layer channel content.** For each writer–reader edge $A \rightarrow B$ supported by path patching, decompose the writer’s matrix

$$W_{OV}^A = \sum_i \sigma_i u_i v_i^\top$$

into rank-one components, following [13]. Use these components to propose candidate communication directions. Static composition scores with the reader’s Q , K , or V projections may guide screening, but are not evidence of a functional channel: in OLMo-2, RoPE makes query–key interactions position-dependent, and QK normalization makes them input-dependent. The separate- Q/K approach suggested in [13] remains an unvalidated extension. Test candidate directions through activation interventions restricted to the writer–reader path, preserving the model’s normalization and positional operations. Measure changes in the reader’s attention or output and in the answer metric, with matched-magnitude control directions. Interpret a direction as a communication channel only when these interventions support its role; weight alignment alone is insufficient.

6. **Divergence-depth skeleton.** Per layer ℓ , the residual-stream distance at the answer position between the token-aligned twins, $d(\ell) = \|h_\ell^{\text{clean}} - h_\ell^{\text{corr}}\|$. The layer where d jumps is where the swap’s consequence reaches the answer position; all head-level findings must be consistent with this curve (a “critical” head above the completed divergence is a contradiction to explain). Nearly free from cached runs; comparison point to the layer-level analyses of parametric multi-hop work.
7. **Minimal circuit and faithfulness.** Define the circuit as the verified nodes + strong edges; run with everything else mean-ablated (means over discovery prompts); **pre-registered criterion:** the circuit must recover $\geq 80\%$ of the full model’s clean–corrupted logit-difference gap on the *validation* families. Completeness check: ablating the circuit from the full model must destroy the behavior.
8. **Generalization.** Repeat the RI scan + exact patching of the identified heads on the *location* world: same heads $\Rightarrow$ general in-context relation machinery; disjoint heads $\Rightarrow$ relation-specific — either outcome is a result.

7 Stage 4 — Quadrant Autopsy

Gate first: group ablations. A single-head zero effect does not prove irrelevance — IOI-style backup heads activate when a primary head is removed [4]. For each false-positive candidate: ablate jointly with all similarly-profiled heads and measure effect vs. coalition size. Only heads inert *in coalition* are declared causally irrelevant; the effect-vs-coalition-size curve doubles as a quantitative distributedness measure.

False positives (high RI, no causal role). What are they instead? Weight fingerprints (stage 1), attention anatomy on our prompts (position? punctuation? relation word? recency?), and behavior on generic copying sequences. Deliverable: an alternative explanation for each head’s high RI — the empirical content behind “the definition over-triggers”.

False negatives (low RI, large effect). What does the definition miss? Expected from our direct-path-blindness analysis: heads whose contribution is indirect (e.g., moving information between fact tokens rather than to the answer position), invisible to a metric that only reads the direct path to the logits. Characterized with the same tools; this quadrant, if populated, is the audit’s sharpest finding.

Afterwards, as time permits: Phase B (two-hop composition, using the same verified single-hop circuit as the component inventory) and/or the checkpoint sweep.

8 Summary of Pre-registered Decisions

Item	Rule
SIH definition parameters	$\tau = 2.2$ exactly; sensitivity $\{1.5, 3.0\}$ in appendix
Head selection threshold	99.9th percentile of the random-triple null (per statistic); sensitivity at 99/99.5
Screen quality gate	Spearman(exact, attribution) ≥ 0.7 on the calibration sample, else upgrade to EAP-IG
Faithfulness criterion	$\geq 80\%$ recovery of the clean–corrupted logit-diff gap on validation families
Audit verdict definitions	sound / incomplete / misleading, as specified in §5.2
Selection data	discovery families only; validation families reserved for faithfulness + generalization

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